Block-masked identity regularisation closes the qft → blocked gap

Setting. DIV2K-8q, QFT(8, 8) basis. `blocked_8` reaches 32.26 dB at $\rho = 0.20$; it is a stable local minimum of QFT(8, 8) (the `qft_warmstart_from_trained_blocked` cell verified this). `qft` (analytic init) reaches 31.29, `qft_identity` (identity init) reaches 31.66. Neither generic-init path crosses to the blocked basin in 1008 steps. Can a structural prior bridge the gap?

Construction

Total loss minimised by the optimiser (Riemannian Adam over $U(2)^{72}$):

$$\mathcal{L}_{\text{total}}(\theta) = \underbrace{\frac{1}{B}\sum_{b=1}^{B}\left\|x_b - T_\theta^\dagger, \text{top}_K(T_\theta\, x_b)\right\|_F^2}_{\text{reconstruction loss (MSE-topK), per minibatch}} + \lambda \cdot \underbrace{R(\theta)}_{\text{regulariser}}$$

Term-by-term:

symbol	meaning
$x_b \in \mathbb{R}^{2^m \times 2^n}$	The b-th image in the minibatch (real, normalised to $[0, 1]$, grayscale, 256×256 for DIV2K/TU-Berlin, 32×32 for QuickDraw).
B	Minibatch size; $B = 50$ for DIV2K/TU-Berlin, $B = 16$ for QuickDraw at the headline preset.
$T_\theta : \mathbb{R}^{2^m \times 2^n} \to \mathbb{C}^{2^m \times 2^n}$	The trainable **forward** QFT(m, n) circuit. Contracts the $(2^m \times 2^n)$ image (reshaped to $(2, 2, ..., 2)$ over $m + n$ qubit axes) through the 72 parametric $U(2)$ gates $\{T_g(\theta)\}$ and reshapes back. For $m = n = 8$, the 72 gates are 16 Hadamards + 56 controlled-phase gates in QFT canonical order.
$T_\theta^\dagger$	The **inverse** circuit. Implemented by conjugating each gate tensor and running the inverse-direction einsum (Hadamards stay Hadamards under conjugation; CPs pick up $-\varphi$). Identical compute cost to T_θ.
$\text{top}_{K(y)}$	**Top-K magnitude truncation**. Sorts the 2^{m+n} entries of $y \in \mathbb{C}^{2^m \times 2^n}$ by $\|y[i, j]\|$, keeps the K largest in-place and zeros the rest. $K = \lfloor 2^{m+n} \cdot 0.1 \rfloor$ at training time — i.e., training is at $\rho = 0.10$ ($K = 6554$ for $m = n = 8$, $K = 102$ for $m = n = 5$); test-time evaluation reports PSNR at $\rho \in \{0.05, 0.10, 0.15, 0.20\}$. Non-smooth at threshold ties, broken stably in implementation.
$\\|M\\|_F^2$	Squared Frobenius norm, $\sum_{i,j} \|M[i, j]\|^2$. The per-image squared reconstruction error, summed over all 2^{m+n} pixels.
λ	Regulariser strength scalar, swept across $\{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ for the block-masked experiment and $\{0.1, 1, 10\}$ for the L1 experiment.
$R(\theta)$	The regulariser — either R_{block} (red equation below) or R_{L1} (red equation in §"L1-to-identity"). Sums a per-gate Frobenius-norm-of-distance-to-identity over the 72 gates, with structural weighting in the block case.

The reconstruction-loss core is: forward-transform the image via T_θ, drop the $1 - \rho$ fraction of smallest-magnitude coefficients (lossy compression), reconstruct via $T_\theta^\dagger$, and measure pixel-wise squared error. The optimiser trains θ to minimise this expected error over the train split **plus** $\lambda \cdot R(\theta)$.

$$R_{\text{block}}(\theta) = \sum_{g \in \mathcal{G}_{\text{outer}}} W \cdot \left\|T_g - I_g\right\|_F^2 + \sum_{g \in \mathcal{G}_{\text{inner}}} \left\|T_g - I_g\right\|_F^2$$

A gate g is **inner** iff every qubit it touches lies in $\{1, 2, 3\}$ (axis 1) or $\{9, 10, 11\}$ (axis 2). Otherwise **outer**. Per-kind identity element: $I_g = I_2$ for Hadamards; $I_g = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ = `controlled_phase_diag(0)` for CPs.

gate kind	inner	outer	total
Hadamards	6	10	16
controlled-phase	6	50	56

gate kind	inner	outer	total
total	**12**	**60**	**72**

The 12/60 split matches `blocked_8`'s sparse shape bit-exactly: 12 trained QFT(3, 3) gates + 60 identity-pinned gates.

Notation

symbol	meaning
θ	72 gate tensors $\in U(2)^{72}$
g	gate label; $\sum_g$ is over the 72 gates
$\mathcal{G}_{\text{inner}}$	12 inner gates
$\mathcal{G}_{\text{outer}}$	60 outer gates ($\mathcal{G}_{\text{inner}} \cap \mathcal{G}_{\text{outer}} = \emptyset$)
$T_g(\theta)$	current 2×2 unitary at gate g
I_g	identity element for gate g's kind (H or CP)
$\|M\|_F$	Frobenius norm: $\sqrt{\sum_{ij} \lvert M_{ij} \rvert^2}$
λ	reg strength; swept $\{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$
W	outer-gate weight; fixed $W = 10$

Key invariant

By construction, at the blocked optimum the outer contribution to R_{block} is zero (every outer gate is bit-exactly at its identity element):

$$R_{\text{block}}\left(\theta_{\text{blocked\_8}}\right) = \sum_{g \in \mathcal{G}_{\text{inner}}} \left\| T_g - I_g \right\|_F^2 + \varepsilon, \quad \varepsilon \approx 0$$

Empirically $R_{\text{block}}\left(\theta_{\text{blocked\_8}}\right) \approx 50.3$ at $W = 10$: inner-sum ≈ 34.5, outer-sum ≈ 1.6 (dominated by one CP outlier; the other 59 outer gates contribute $< 10^{-2}$ each). The reg term does **not** push the operator away from blocked even at large λ — this is the design property that enables crank-up without collapse.

Implementation

`pdft_benchmarks.identity_reg.BlockMaskedIdentityRegQFTMSELoss`, via the `MSELoss._extra_loss` hook (pdft PR #18). Headline preset: 1008 steps, batch 50, val split 0.15, seed 42, `--no-early-stop`. Start from `qft_identity` init.

Result

Test PSNR (dB) after 1008 steps:

basis / reg	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
`qft` (analytic init)	25.09	27.57	29.53	31.29
`qft_identity` (identity init)	25.23	27.81	29.84	31.66
`qft_identity` + reg, $\lambda = 10^{-3}$	25.23	27.81	29.84	31.66
`qft_identity` + reg, $\lambda = 10^{-2}$	25.23	27.81	29.84	31.66
`qft_identity` + reg, $\lambda = 10^{-1}$	25.24	27.81	29.84	31.66
`qft_identity` + reg, $\boldsymbol{\lambda = 1}$	**25.18**	**28.09**	**30.30**	**32.26**
`qft_identity` + reg, $\lambda = 10$	25.18	28.08	30.29	32.25
`blocked_8` (warm-start ceiling)	25.18	28.09	30.30	32.26

At $\lambda = 1$, the trained operator matches `blocked_8` **bit-exactly to 2 decimals at every $\boldsymbol{\rho}$**. Trajectory crosses from 31.66 dB (qft_identity basin) to 32.26 dB (blocked basin), closing the full 0.60 dB gap. $\lambda = 10$ within 0.01 dB.

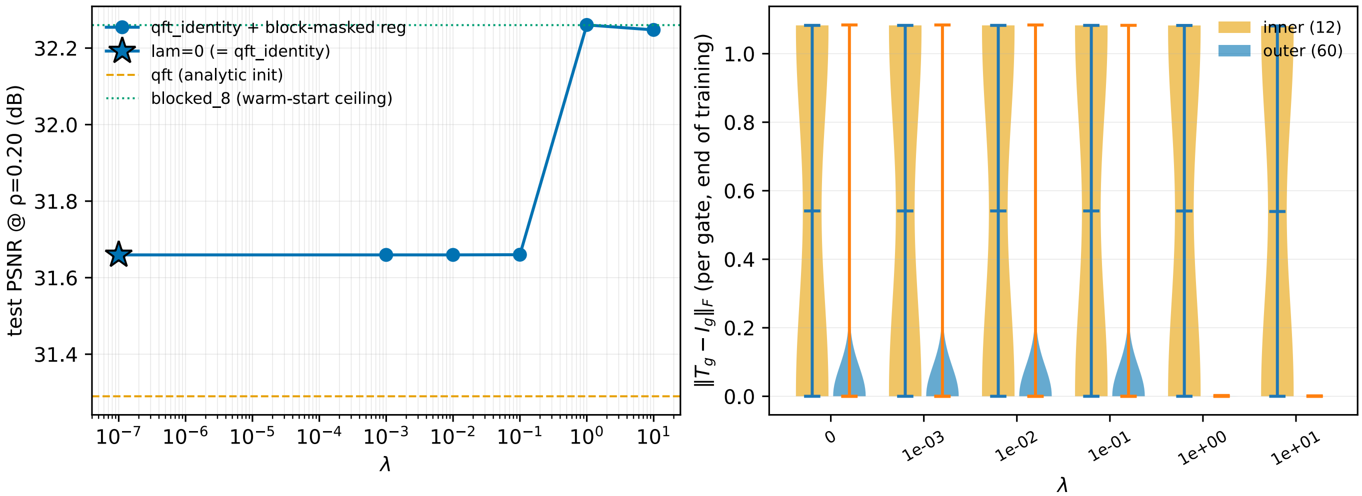

Figure 1: Left: test PSNR at $\rho = 0.20$ vs λ (log x). Sharp transition $31.66 \to 32.26$ dB between $\lambda = 10^{-1}$ and $\lambda = 1$. Right: per-gate $\left\|T_g - I_g\right\|_F$ at end of training, inner (12, orange) vs outer (60, blue). At $\lambda \geq 1$, outer gates collapse to a thin line at zero — **all 60 pinned bit-exactly at identity**, matching `blocked_8`′s structural shape. Inner-gate distributions are invariant across λ: the prior leaves them free.

Interpretation

Outcome (i) from the spec's three-falsifiable-outcomes frame: the blocked basin was **unfavored, not isolated**. Given the matched structural prior, smooth-Riemannian Adam reaches it from identity init in 1008 steps with no curriculum and no warm-start.

Leading-prior caveat. The regulariser bakes in `blocked_8`′s known structural shape. A positive result is "if you tell the optimiser the answer-shape, it finds the answer" — closer to warm-start with extra steps than to basin discovery from scratch.

The shift in framing: the headline `qft` < `blocked_8` gap is now attributable to **initialisation + optimiser-coupling**, not to a loss-landscape obstruction. The remaining open question is **how weak a prior suffices**.

L1-to-identity: dropping the block-aligned mask

Drop the inner/outer mask; penalise all gates uniformly with an L1 norm (Huber-smoothed to avoid the kink at $T_g = I_g$):

$$R_{\mathrm{L1}}(\theta) = \sum_g \sqrt{\left\|T_g - I_g\right\|_F^2 + \varepsilon}, \quad \varepsilon = 10^{-12}$$

The unsmoothed L1's gradient $\frac{T_g - I_g}{\|T_g - I_g\|_F}$ is $\frac{0}{0}$ at qft_identity init; `jax.grad` returns NaN there. The ε-smoothing yields a finite ($= 0$) gradient at the kink while agreeing with true L1 to leading order for $\left\|T_g - I_g\right\|_F \gtrsim 10^{-5}$. L1 induces sparsity from training dynamics: gates either move appreciably or stay pinned; no continuous shrinkage.

L1 sweep — DIV2K-8q. $\lambda \in \{0.1, 1, 10\}$, qft_identity init, same headline preset:

basis / reg	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$	active gates
`block_dct_8` (classical)	**26.11**	**29.41**	**31.86**	**34.01**	—
`qft_identity` (no reg)	25.23	27.81	29.84	31.66	72/72
L1 reg, $\lambda = 0.1$	25.23	27.81	29.84	31.66	72/72
L1 reg, $\lambda = 1$	25.23	27.81	29.84	31.66	72/72
L1 reg, $\lambda = 10$	25.18	28.08	30.30	32.26	6/72
`blocked_8` (ceiling)	25.18	28.09	30.30	32.26	12/72

L1 at $\lambda = 10$ matches `blocked_8` to ≈ 0.02 dB at every ρ, with **only 6 active gates** (vs `blocked_8`′s 12). The 6 are exactly the inner-block Hadamards (indices $0, 1, 2, 8, 9, 10$); all 6 inner CPs that `blocked_8` makes non-trivial are pinned at identity by L1. The 6 active gates trained to rotation-by-$\frac{\pi}{4}$ matrices $\begin{pmatrix} c & s \\ -s & c \end{pmatrix}$ with $c, s \approx \frac{1}{\sqrt{2}}$ — **not** Hadamards $\begin{pmatrix} c & s \\ s & -c \end{pmatrix}$. The basin is structurally distinct from `blocked_8` (no CP entanglement) yet PSNR-equivalent.

Comparison against the full block-size grid (DIV2K-8q):

basis	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
blocked_4	19.06	26.93	29.76	32.05
blocked_8	**25.18**	**28.09**	**30.30**	**32.26**
blocked_16	25.23	27.81	29.84	31.66
blocked_32	24.91	27.30	29.20	30.91
L1 $\lambda = 10$	**25.18**	**28.08**	**30.30**	**32.26**

L1 hits the **training-rate-optimal** blocked ceiling at every ρ (blocked_8 wins at $\rho \geq 0.10$; at $\rho = 0.05$ blocked_16 wins by 0.05 dB and L1 matches blocked_8 rather than blocked_16). Loss trains at $\rho = 0.10$ via $k = \lfloor 2^{m+n} \cdot 0.1 \rceil$.

L1 sweep — QuickDraw ($m = n = 5$). Same procedure, 30-gate QFT(5, 5) basis:

basis / reg	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$	active gates
`block_dct_8` (classical)	17.20	20.70	23.72	26.63	—
`qft` (analytic init)	16.72	19.58	22.05	24.36	30/30
blocked_4 / blocked_8	18.12	22.40	26.18	30.04	varies
L1 reg, $\lambda = 0.1$	17.24	23.33	30.06	40.62	2/30
L1 reg, $\lambda = 1$	12.82	18.35	30.48	54.11	0/30
L1 reg, $\lambda = 10$	**12.82**	**18.34**	**30.85**	**61.89**	**0/30**

On QuickDraw, L1 finds a **different** answer: at $\lambda \geq 1$ **every gate is pinned at identity** and the trained operator is literally $T(x) = x$. PSNR at $\rho = 0.20$ reaches 61.89 dB — 30 + dB above any `blocked_b` and 37 + dB above analytic `qft`. QuickDraw images are sparse line drawings; top-k truncation in the pixel domain already captures essentially all the strokes. A transform basis is **worse** than no transform.

Caveat. The 61.89 dB QuickDraw number is partly a property of the data, not just the regulariser: at 32×32 with mostly-zero pixels, top-20%-of-pixels retention is near-perfect by construction. The DIV2K result is the more discriminating signal because DIV2K is not pixel-sparse.

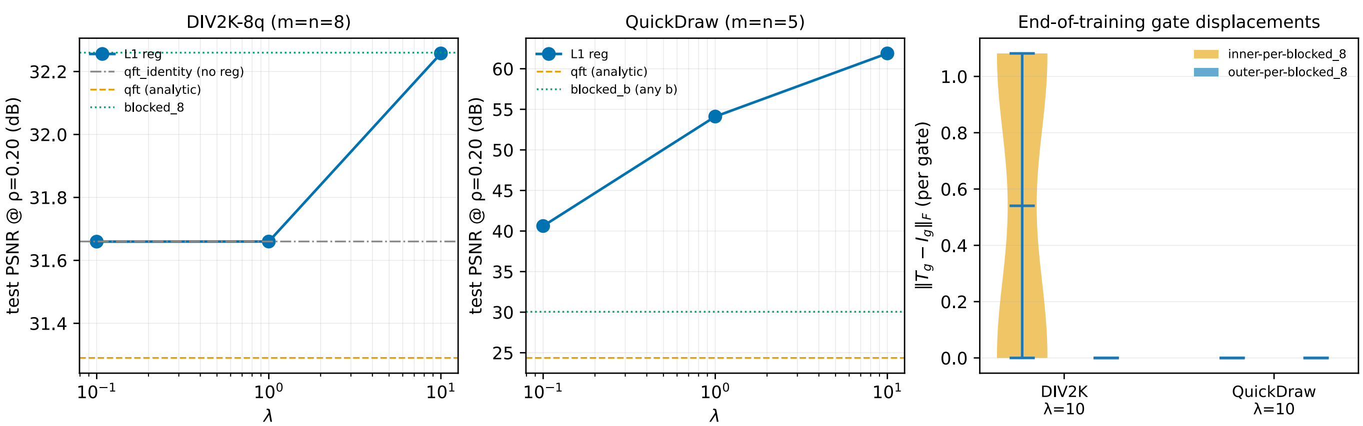

Figure 2: Left, middle: test PSNR at $\rho = 0.20$ vs λ for L1-regularised qft_identity on DIV2K-8q and QuickDraw. On DIV2K, L1 matches `blocked_8` at $\lambda = 10$; on QuickDraw L1 **exceeds** every `blocked_b` by 30 + dB at $\lambda \geq 1$. Right: end-of-training per-gate $\left\|T_g - I_g\right\|_F$ at $\lambda = 10$ for both datasets, coloured by `blocked_8`'s inner (12, orange) / outer (60, blue) mask. DIV2K: 6 inner-Hadamard gates are active at distance ≈ 1.08; everything else is bit-exactly at identity. QuickDraw: **all** gates are bit-exactly at identity — L1 picked no transform.

Frequency-space evidence

Per-image frequency representations $|T(x)|$ and reconstructions at $\rho = 0.20$ make the dataset-adaptive behaviour visible.

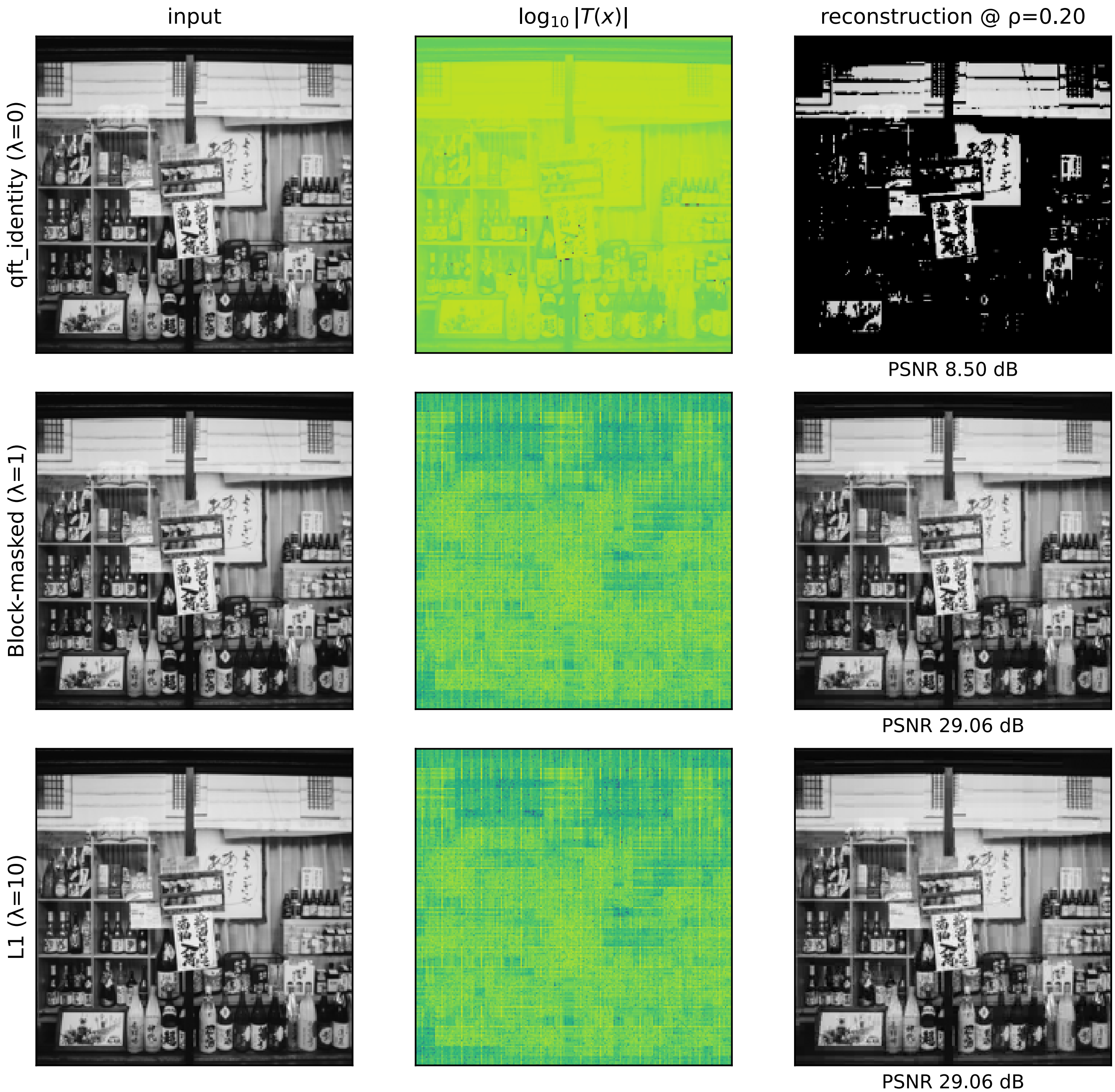

Figure 3: **DIV2K-8q, single test image.** Rows: `qft_identity` at bare init (identity operator, no training), block-masked $\lambda = 1$ (trained), L1 $\lambda = 10$ (trained). Columns: input, $\log_{10}|T(x)|$, reconstruction at $\rho = 0.20$. The identity operator (top row) leaves the image in pixel domain; top-20% pixel retention destroys it (PSNR 8.50 dB on this image). Both trained operators (rows 2 and 3) produce structured frequency spectra and recover the image cleanly. Block-masked and L1 trained operators yield visibly similar $|T(x)|$ panels despite training under different regulariser shapes — consistent with both landing on PSNR-equivalent points of the QFT(8, 8) loss surface.

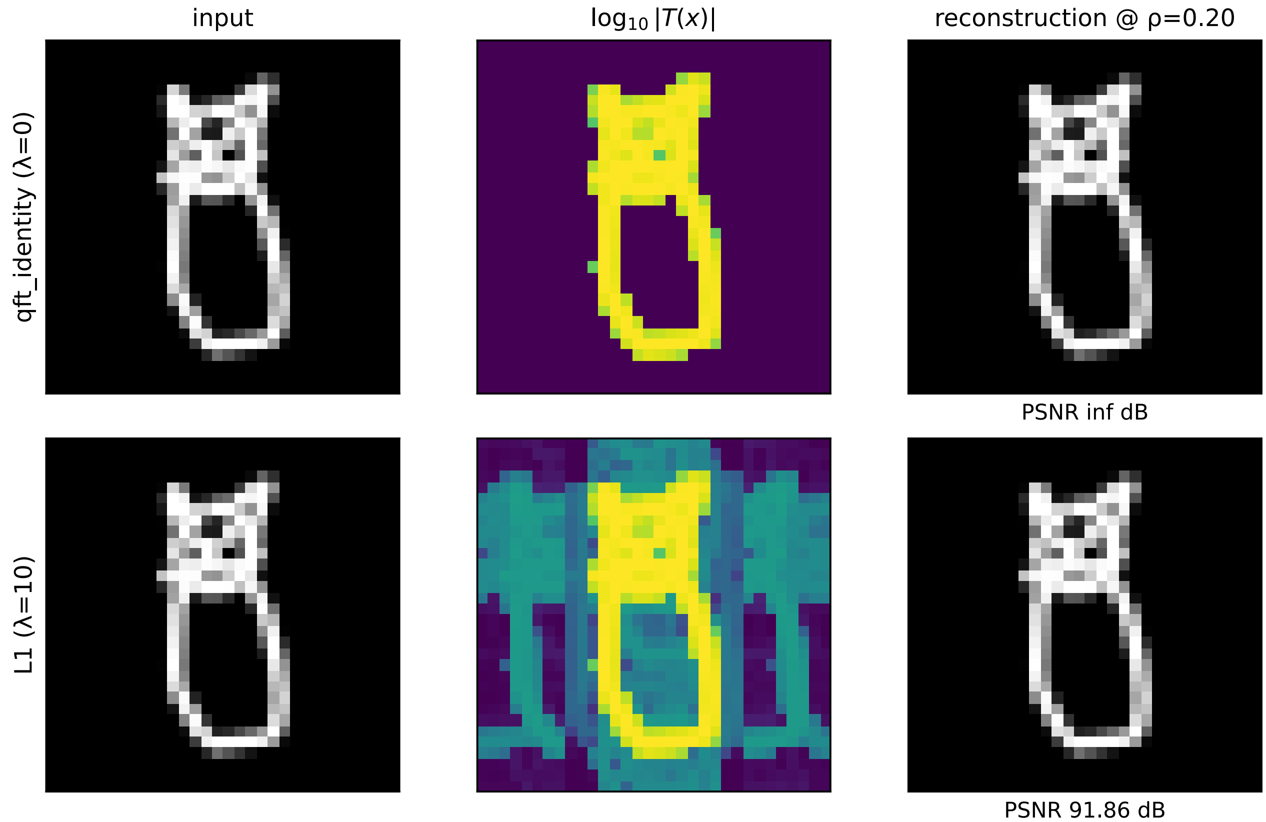

Figure 4: **QuickDraw, single test image.** Rows: `qft_identity` at bare init, L1 $\lambda = 10$ (trained). The L1-trained operator is **essentially the identity** — its frequency representation nearly matches the input (only faint perturbations on the column-sum lines). On pixel-sparse data, top-20% retention of the raw pixels reconstructs the image to numerical precision (PSNR ∞ at init, 91.86 dB after L1 training — both effectively perfect). The transform was unnecessary.

L1 sweep — TU-Berlin ($m = n = 8$). Larger sketch dataset at the DIV2K scale (256×256), 20{,}000 hand-drawn sketches across 250 categories (CC-BY-4.0; HuggingFace mirror `sdiaeyu6n/tu-berlin`). Same QFT(8, 8) basis, same headline preset; tests whether the QuickDraw "L1 picks identity" finding is a 32×32 pixel-sparsity artefact or a genuine pixel-sparse-data phenomenon:

basis / reg	$\rho = 0.05$	$\rho = 0.10$	$\rho = 0.15$	$\rho = 0.20$
`block_dct_8` (classical)	41.31	61.88	90.70	117.58
L1 reg, $\lambda = 10$	68.69	104.54	114.28	120.47
L1 reg, $\lambda = 1$	68.48	104.72	115.32	121.77
L1 reg, $\lambda = 0.1$	**68.78**	**105.38**	**116.04**	**123.08**

All three L1 λ values reach near-exact reconstruction (> 120 dB at $\rho = 0.20$), confirming the QuickDraw finding is **not** a 32×32 artefact: pixel-sparse content at 256×256 also resolves under L1 into a near-identity operator. `block_dct_8` trails L1 by 5.5 dB at $\rho = 0.20$ — block-DCT transforms the already-sparse pixel signal **away** from its native sparsity.

The λ dependence is **inverted** relative to QuickDraw: smaller $\lambda \to$ slightly higher PSNR (L1 $\lambda = 0.1$ wins at $\rho = 0.20$, 123.08 dB). At $m = n = 8$, 72 gates have more room to perturb without hurting; a tiny amount of structure beyond pure identity helps marginally on TU-Berlin.

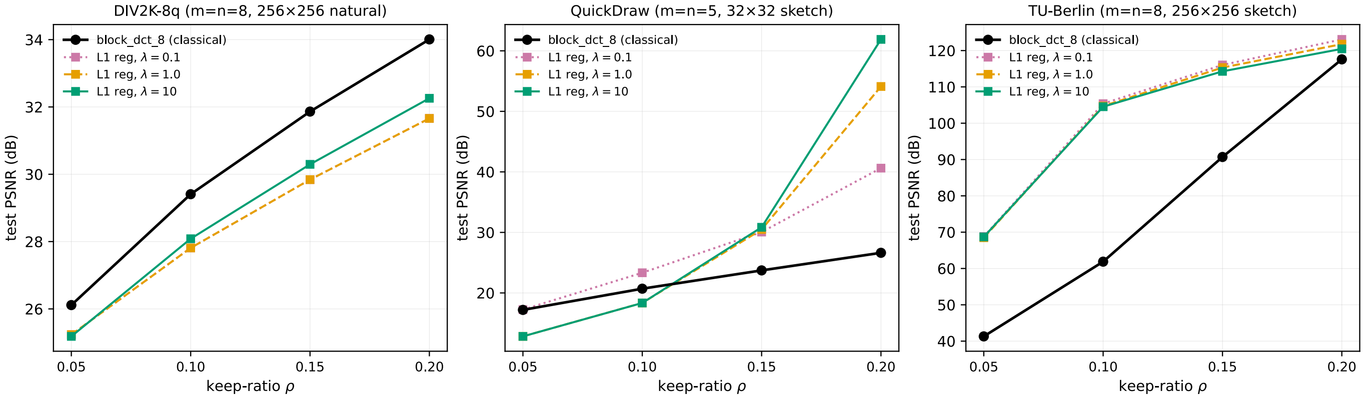

Figure 5: **Cross-dataset PSNR(ρ) summary.** Each panel: PSNR vs keep-ratio ρ for one dataset. Black solid: `block_dct_8` (classical JPEG-style 8×8 block DCT, no training). Coloured squares: L1 reg at $\lambda \in \{0.1, 1, 10\}$. **Three regimes:** (left, DIV2K-8q natural images) **`block_dct_8` wins** by 1.75 dB — JPEG-style block-DCT remains the empirical optimum on natural-image content at $\rho = 0.20$; L1′s 6-rotation basin trails. (middle, QuickDraw 32×32 sketch) L1 $\lambda = 10$ **dominates** at high ρ by 35 + dB — identity-basis L1 beats block-DCT massively because the data is already pixel-sparse. (right, TU-Berlin 256×256 sketch) all three L1 λ values dominate `block_dct_8` by 5.5 dB at $\rho = 0.20$ and by larger gaps at lower ρ. The sketch- dataset finding replicates at the DIV2K spatial scale, confirming it is not a low-resolution pixel-sparsity artefact.

What this reveals

L1 is **not** finding `blocked_8`. It is finding the **best basis** for the dataset under the sparsity prior:

- DIV2K (transform-needy): L1 picks 6 rotations on the inner-block qubits. PSNR equal to the training-rate-optimal `blocked_b`, structurally simpler (6 gates vs 12, no CP entanglement).
- QuickDraw (already pixel-sparse): L1 picks the identity basis. PSNR far above any `blocked_b`, because no transform was needed.

This is substantially stronger than the original "auto-discover block size" framing: L1 generalises beyond the `blocked_b` family entirely. The block-aligned mask is one valid sparse-basis shape; L1 finds others when they exist (DIV2K's no-CP basin) or returns to identity when no transform is needed (QuickDraw).

The headline `qft` < `blocked_8` gap is then explained as **Adam-from-qft__identity is poorly coupled to the sparse local minima of the QFT(8, 8) loss surface**. Any sparsity-inducing prior — matched mask (block-aware L2), generic sparsity (L1), or even the warm-start init — steers training to one of those minima.

Reproducibility

```
python experiments/qft_identity_regularization.py \
    --gpu 0 --reg block --lambdas 0,1e-3,1e-2,1e-1,1,10 \
    --outer-weight 10 --epochs 112
```

Per-run output: `results/qft_identity_init/div2k_8q/_runs/reg_lambda_<lam>_W<W>/`. Run time: ≈ 9 min per λ on RTX 3090. 17 unit tests in `tests/test_identity_reg.py`.